

Ethiopian Birr Banknote Denomination Recognition

Capstone Project Summary

Using Deep Learning and Transfer Learning

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1. Problem Definition

Currency recognition is a fundamental task in financial transactions, retail automation, and accessibility tools for visually impaired individuals. In Ethiopia, banknotes of different denominations share similar physical dimensions and color characteristics, making manual identification challenging - particularly for people with visual impairments.

Currently, there is a lack of automated, accessible tools specifically designed to recognize Ethiopian Birr banknote denominations. Existing currency recognition systems are primarily built for widely circulated currencies such as USD, EUR, and INR, leaving a significant gap for Ethiopian currency users.

This project addresses the need for an intelligent system that can automatically identify Ethiopian Birr banknote denominations (5, 10, 50, 100, and 200 Birr) from images. The system uses deep learning - specifically Convolutional Neural Networks (CNNs) and transfer learning - to classify banknote images captured via camera or uploaded by the user.

The solution is deployed as a web application with voice output, making it practical and accessible for real-world use, including assistive technology for visually impaired users.

2. Objectives

General Objective

To develop a deep learning-based system that accurately recognizes and classifies Ethiopian Birr banknote denominations from images, and to deploy it as an accessible web application.

Specific Objectives

- Collect and preprocess a dataset of Ethiopian Birr banknote images covering five denominations.
- Build and train a baseline CNN model for multi-class image classification.
- Apply transfer learning using MobileNetV2 to improve classification accuracy.
- Compare the performance of the custom CNN versus the transfer learning approach.
- Deploy the trained model as an interactive web application using Streamlit.
- Implement voice output (text-to-speech) for accessibility.

3. Dataset Description

The dataset consists of 1,919 images of Ethiopian Birr banknotes, organized into 11 sub-classes (front and back of each denomination, plus a background class). The images were collected under varying lighting conditions and angles to improve model robustness.

For model training, front and back images of each denomination are mapped to the same class label, resulting in 5 denomination classes: 5 Birr, 10 Birr, 50 Birr, 100 Birr, and 200 Birr.

Dataset Distribution

Denomination	Train (F)	Train (B)	Test (F)	Test (B)	Total
5 Birr	148	148	36	36	368
10 Birr	148	148	36	36	368
50 Birr	148	148	36	36	368

100 Birr	148	148	36	36	368
200 Birr	148	148	36	36	368
Background	64	-	15	-	79

The dataset is balanced across the five denomination classes, with 148 training and 36 test images per side per denomination. A validation set of 20% was created from the training data.

Data Preprocessing

- Images resized to 224 x 224 pixels to match the MobileNetV2 input requirements.
- Pixel values normalized from [0, 255] to [0, 1] range.
- 80/20 train-validation split applied to the training set.

Data Augmentation

To reduce overfitting and increase data diversity, the following augmentation techniques were applied during training:

- Random rotation (up to 5 degrees)
- Random zoom (up to 10%)
- Random horizontal and vertical translation (up to 10%)

4. Methodology

The project follows a systematic deep learning workflow consisting of the following phases:

Phase 1: Data Collection and Exploration

Images of Ethiopian Birr banknotes were collected and organized into labeled directories. Exploratory Data Analysis (EDA) was performed to verify class distributions, visualize sample images, and ensure data quality.

Phase 2: Preprocessing

All images were resized to 224x224 pixels and normalized. Labels were encoded as integers (0-4) for sparse categorical cross-entropy loss. A validation set was created from the training data to monitor overfitting.

Phase 3: Model Development

Two models were developed and compared:

- A custom CNN built from scratch (baseline model).
- A MobileNetV2-based transfer learning model (final model).

Both models were trained with data augmentation and evaluated on the held-out test set.

Phase 4: Evaluation

Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices to assess overall and per-class performance.

Phase 5: Deployment

The best-performing model (MobileNetV2) was saved and integrated into a Streamlit web application with camera input, image upload, and voice output capabilities.

5. Model Architecture

5.1 Baseline CNN

The baseline model is a custom Sequential CNN with four convolutional blocks, followed by a Global Average Pooling layer and a dense classifier. Each convolutional block consists of:

- Conv2D layer with increasing filters (32, 64, 128, 256)
- Batch Normalization
- ReLU Activation
- MaxPooling2D (2x2)

The classification head includes a 128-unit Dense layer with ReLU, a Dropout layer (0.3), and a 5-unit Dense layer with Softmax activation.

5.2 MobileNetV2 Transfer Learning (Final Model)

The final model leverages MobileNetV2, a lightweight and efficient CNN pre-trained on the ImageNet dataset (1.4 million images, 1,000 classes). The architecture consists of:

- Data augmentation layer (rotation, zoom, translation)
- Rescaling layer to convert pixel values to [-1, 1] range (MobileNetV2 requirement)

- Frozen MobileNetV2 base (feature extractor, not trainable)
- Global Average Pooling 2D
- Dense layer (128 units, ReLU activation)
- Dropout (0.3)
- Output Dense layer (5 units, Softmax activation)

Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	0.001 (with ReduceLROnPlateau)
Loss Function	Sparse Categorical Cross-Entropy
Batch Size	32
Epochs	20 (with EarlyStopping, patience=5)
Input Size	224 x 224 x 3
Validation Split	20% of training data

6. Training and Evaluation

6.1 Baseline CNN Training

The baseline CNN was trained for 20 epochs. While training accuracy climbed to ~90%, the validation accuracy remained low (~25-44%), indicating severe overfitting. The model memorized training patterns but failed to generalize to unseen data.

6.2 MobileNetV2 Training

The MobileNetV2 transfer learning model converged much faster and showed strong generalization. Key observations:

- Validation accuracy reached ~91% by epoch 8.
- ReduceLROnPlateau reduced the learning rate when validation loss plateaued.
- EarlyStopping restored the best weights to prevent overfitting.
- The gap between training and validation accuracy remained small, indicating good generalization.

7. Results

7.1 Model Comparison

Metric	Baseline CNN	MobileNetV2
Test Accuracy	64%	94%
Test Loss	0.951	0.165
Overfitting	Severe	Minimal
Training Time	~13 min	~4 min

7.2 MobileNetV2 Classification Report

Class	Precision	Recall	F1-Score	Support
5 Birr	0.94	0.90	0.92	72
10 Birr	0.96	0.97	0.97	72
50 Birr	0.96	0.97	0.97	72
100 Birr	0.91	0.93	0.92	72
200 Birr	0.94	0.93	0.94	72
Overall	0.94	0.94	0.94	360

The MobileNetV2 model achieved 94% overall accuracy on the test set, with consistently high precision, recall, and F1-score across all five denomination classes. The 10 Birr and 50 Birr classes achieved the highest performance (F1 = 0.97), while 5 Birr and 100 Birr showed slightly lower but still strong results (F1 = 0.92).

7.3 Baseline CNN Classification Report

Class	Precision	Recall	F1-Score	Support
5 Birr	0.67	0.54	0.60	72
10 Birr	0.88	0.53	0.66	72

50 Birr	0.42	0.96	0.58	72
100 Birr	0.79	0.58	0.67	72
200 Birr	1.00	0.58	0.74	72
Overall	0.75	0.64	0.65	360

The baseline CNN performed significantly worse, with only 64% accuracy and severe class imbalance in predictions - the 50 Birr class dominated predictions (recall = 0.96 but precision = 0.42), while other classes had low recall.

8. Web Application

The trained MobileNetV2 model is deployed through a Streamlit web application that provides an intuitive interface for banknote recognition. The application is designed with accessibility in mind, featuring automatic voice output of prediction results.

Application Features

- Image Upload: Users can upload banknote images in JPG, JPEG, or PNG format.
- Camera Capture: Users can take photos directly from their device camera.
- Real-time Prediction: The model processes the image and displays results instantly.
- Confidence Score: Shows prediction confidence as a percentage.
- Probability Breakdown: Displays per-class probabilities with progress bars.
- Voice Output: Automatically speaks the predicted denomination using the Web Speech API.
- Confidence Alerts: Provides feedback based on confidence level - high (>90%), moderate (70-90%), or low (<70%).

How It Works

1. The user selects an input method (upload or camera).
2. The uploaded/captured image is converted to RGB and resized to 224x224 pixels.
3. Pixel values are normalized to the [0, 1] range.
4. The image is fed through the MobileNetV2 model for prediction.
5. The predicted class (denomination), confidence, and probability distribution are displayed, and the result is spoken aloud.

Running the Application

```
streamlit run app.py
```

9. Technologies Used

Category	Technology
Programming Language	Python 3.10+
Deep Learning	TensorFlow / Keras
Pre-trained Model	MobileNetV2 (ImageNet)
Web Framework	Streamlit
Image Processing	Pillow (PIL), NumPy
Data Analysis	Pandas, Matplotlib, Seaborn
Machine Learning	Scikit-learn
Voice Output	Web Speech API (browser-native)
Development	Jupyter Notebook
Version Control	Git / GitHub

10. Conclusion and Future Work

Conclusion

This project successfully demonstrates that deep learning, particularly transfer learning with MobileNetV2, can effectively classify Ethiopian Birr banknote denominations with high accuracy (94%). The comparison between the custom CNN (64% accuracy) and the MobileNetV2 model clearly shows the advantage of leveraging pre-trained features from large-scale datasets like ImageNet.

The deployed Streamlit web application provides a practical, user-friendly interface that supports both image upload and camera capture, with voice output for accessibility. The system is lightweight and can run on standard hardware without GPU requirements.

Key Achievements

- Built and compared two deep learning models for Ethiopian Birr denomination recognition.
- Achieved 94% test accuracy using MobileNetV2 transfer learning.
- Deployed a fully functional web application with camera and voice capabilities.
- Created an accessible tool that can assist visually impaired users.

Limitations

- The dataset size is relatively small (1,919 images), which may limit generalization.
- The model only recognizes five denominations and does not detect counterfeit notes.
- Performance may degrade under poor lighting conditions or extreme angles.
- The voice output relies on browser-native Web Speech API, which may vary across browsers.

Future Work

- Expand the dataset with more diverse images (different lighting, backgrounds, wear conditions, and partially occluded notes).
- Add support for the 1 Birr note and coin denominations.
- Implement counterfeit detection as an additional classification layer.
- Fine-tune the MobileNetV2 backbone (unfreeze later layers) for potentially higher accuracy.
- Deploy the application as a mobile app for greater accessibility.
- Add Amharic language support for the voice output and user interface.
- Explore model optimization (quantization, TFLite) for on-device inference.

References

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